

LifeLog

Unified Personal Activity Intelligence Database

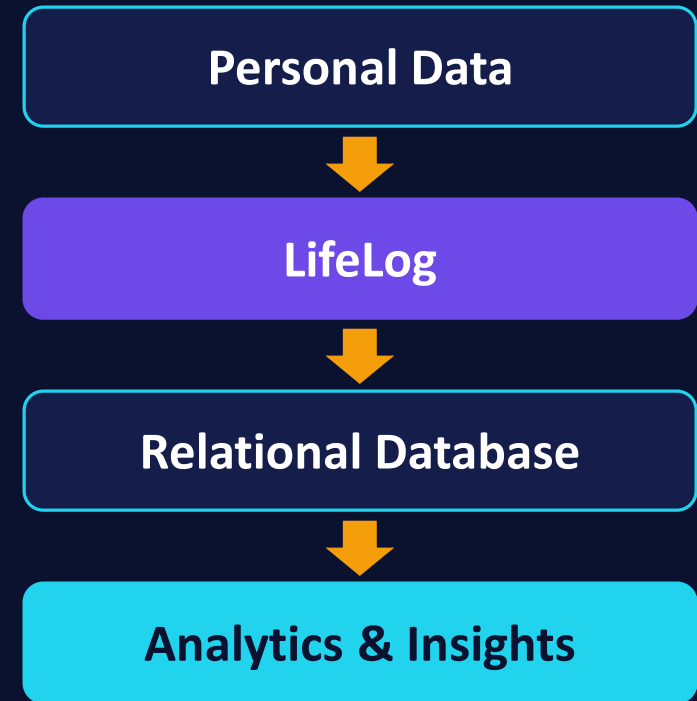
“Turning everyday personal activity into meaningful, connected insight”

Haadi Hasil	2520030297
Jaiswal Reddy	2520030135
Raghava Akkinepally	2520030271

Guide: Dr. Y Prasanthi

Assistant Professor, Department of Computer Science and Engineering
Koneru Lakshmaiah Education Foundation

Database Systems Engineering and Distributed Backend Development · PBL 2026–27



Study



Expenses



Food



Sleep



Mood



Transport



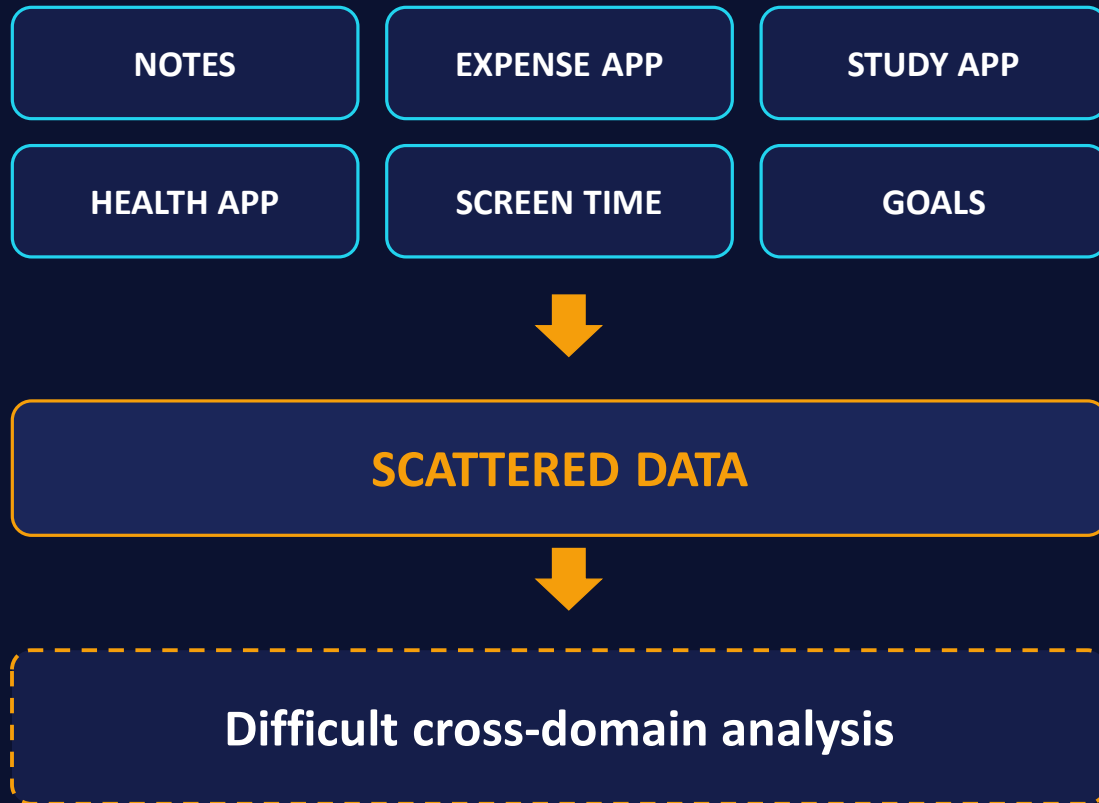
Screen Time



Goals

The Problem: Personal Data Exists in Silos

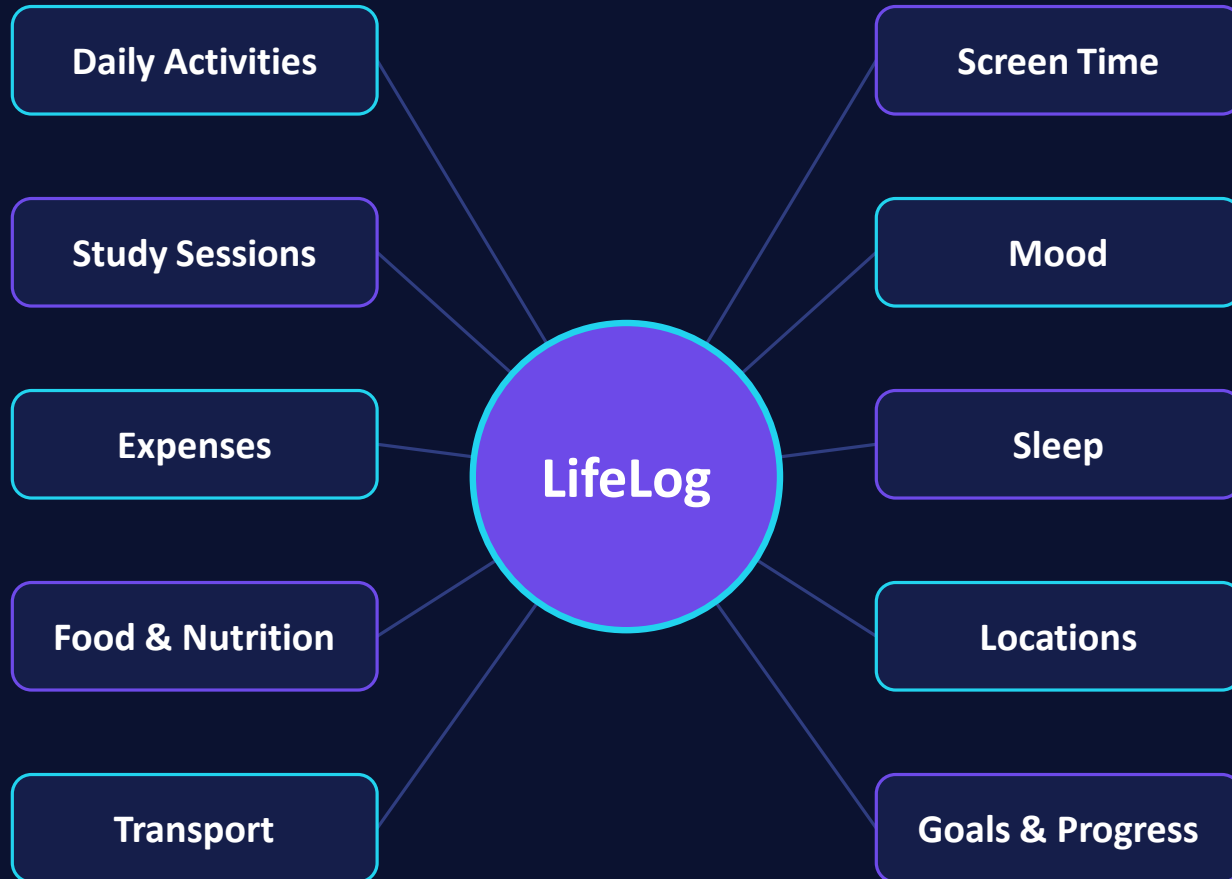
Every day we generate: what we studied · spent · ate · how we travelled · screen time · sleep · mood · goals



- 1 Data is scattered across notebooks and unrelated apps
- 2 Different domains share no common data model
- 3 Links between sleep, mood, study and spending are hard to analyze
- 4 Trends and summaries need manual calculation
- 5 Historical retrieval is slow and tedious
- 6 Data may be lost, duplicated or inconsistently recorded

LifeLog brings these domains together using one normalized relational database.

LifeLog — Proposed Solution



LifeLog is a unified personal activity tracking and analytics system built on a single relational database.

Personal records are stored as structured, validated data owned by an authenticated user, connected through a common relational model.

Objectives

- Design a normalized relational database
- Record personal activities reliably
- Track sleep, mood and screen time
- Manage personal goals and progress
- Enforce integrity with keys, constraints, triggers
- Summarize via views, procedures, aggregates
- Expose operations through a Spring Boot REST API
- Visualize trends in an analytics dashboard

System Architecture



Database Design

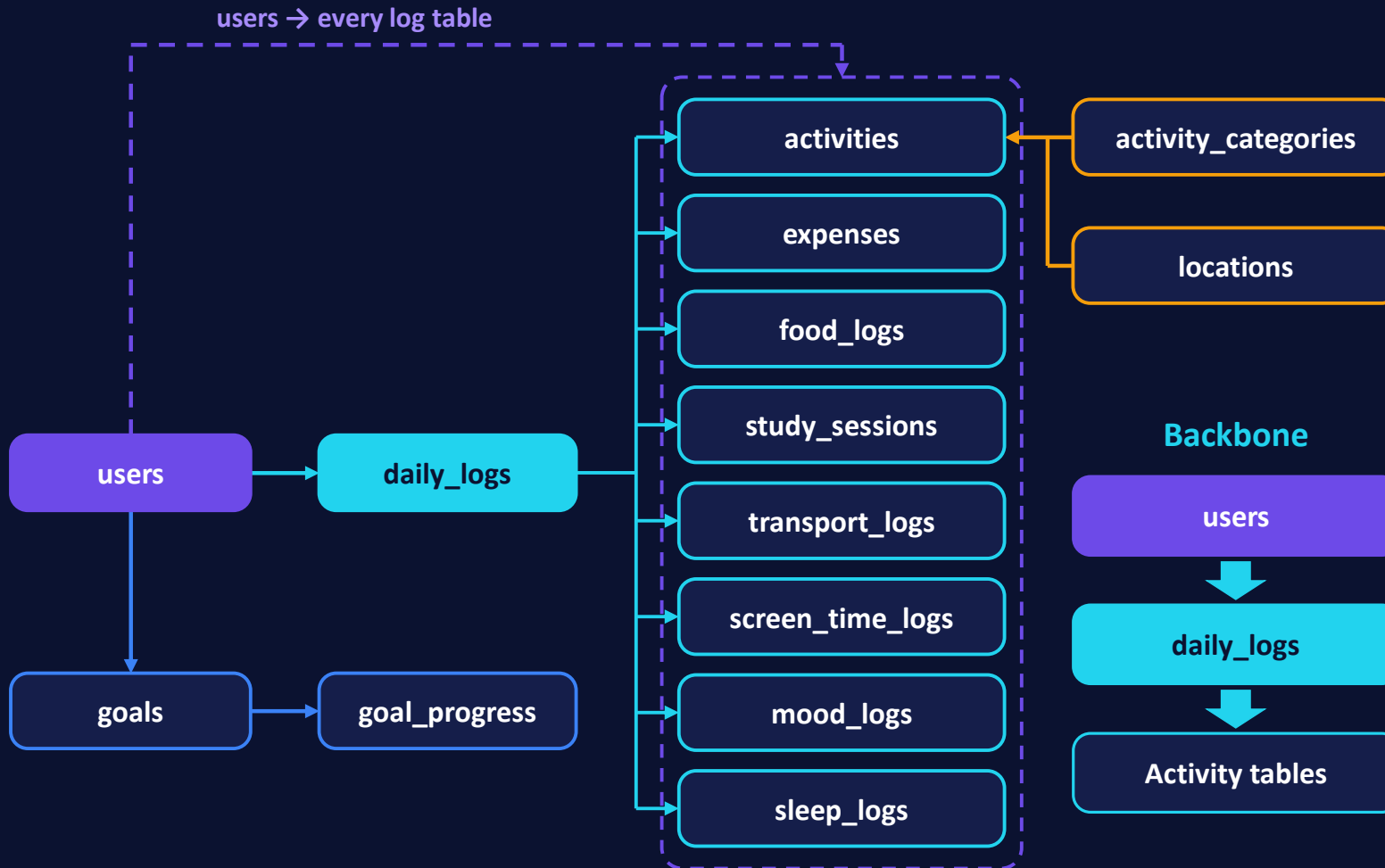
The lifelog_db schema contains 14 related tables.

14 TABLES

3NF

InnoDB

RELATIONAL MODEL



Database characteristics

- 3NF
- InnoDB
- Primary Keys
- Foreign Keys
- Composite Keys where required
- Referential Integrity
- Cascading rules
- Trigger-updated goal progress

Implementation & Technology Stack



FRONTEND

- HTML
- CSS
- JavaScript
- Fetch API
- Chart.js



BACKEND

- Java 17
- Spring Boot 3.x
- Spring Data JPA
- REST APIs



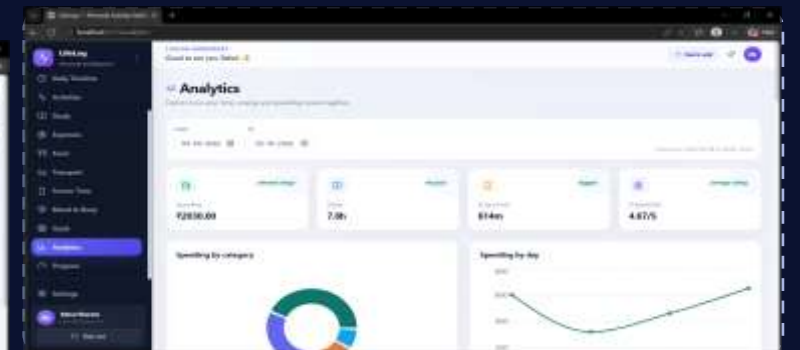
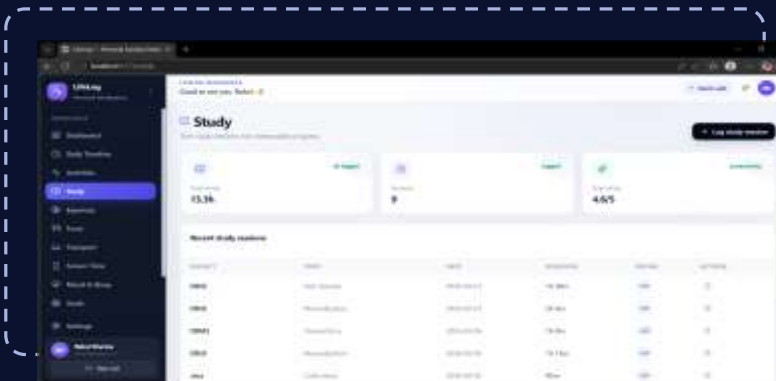
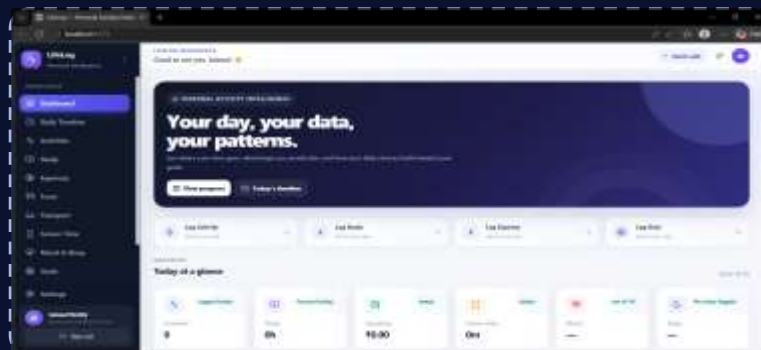
DATABASE

- MySQL 8+
- InnoDB
- SQL
- Views
- Stored Procedures
- Triggers
- Transactions



DEVELOPMENT

- Maven
- Git
- GitHub
- Web Browser
- Windows / Linux



Results, Testing & Feasibility

Results

- Activity data centralized in one relational system
- 14 related tables form the database model
- Major activity domains stored as structured records
- Joins and aggregation support cross-domain analysis
- Views and procedures give reusable operations
- Triggers and constraints support data integrity
- Transactions keep multi-step operations consistent
- REST architecture gives a structured interface

Feasibility

TECHNICAL

Java + Spring Boot + MySQL stack

ECONOMIC

Open-source development technologies

OPERATIONAL

Browser-based dashboard

SCHEDULE

Semester PBL implementation

Testing / Validation

Test Area	Expected Result
CRUD Operations	Records can be created, retrieved, updated and deleted
Foreign Keys	Invalid relationships are prevented
Constraints	Invalid data is rejected
Joins	Related records can be retrieved
Aggregations	Summaries can be generated
Views	Reusable analytical queries work
Triggers	Automated integrity/progress actions execute
Transactions	Multi-step operations maintain consistency

Conclusion & Future Scope

“LifeLog brings scattered personal activity data into one normalized relational database and makes it queryable.”

Key achievements

Unified relational data model

14 related tables

Normalization up to 3NF

REST-based backend architecture

SQL-based analytics

Data integrity mechanisms

Dashboard-based visualization

Practical use of DBMS concepts

From fragmented personal records to one connected data system.

Future scope

- Android and iOS companion application
- Reminders for logs and goal deadlines
- Fitness and banking integrations
- Smarter habit recommendations
- Shared goals with friends or mentors
- PDF and CSV report export
- Cloud-hosted database with backups
- Two-factor authentication

THANK YOU

Questions?

Team: Haadi Hasil | Jaiswal Reddy | Raghava Akkinapally
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